

Good Titration Practice



Basic User
Familiarization
Course
October 2012

METTLER TOLEDO

Agenda



Sources of errors

- Using the right titration method
- Reagent handling
- pH calibration and sensor care
- Sampling & Weighing the right way
- Practicals

Sources of Error



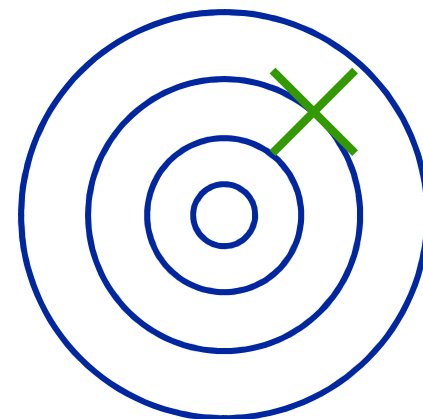
Systematic error:

a generally unknown component of the error which remains constant or drifts slightly

Examples:

- Different analytical method
- Wrong sample size
- Titrant concentration
- Blank value
- Sensor adjustment/calibration
- Sensor response
- Sensor conditioning

Inaccurate result



Sources of Error



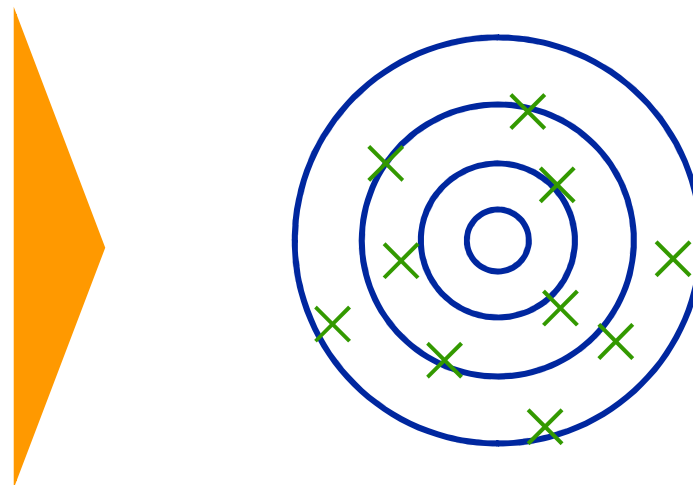
Random error:

a component of the error which varies in an unpredictable way

Examples:

- Inadequate equipment
- Method parameters
- Bubbles
- Sample handling
- Sample
- Rinsing
- Operator

Imprecise result



Sources of Error



How to distinguish: **Systematic vs. Random?**

REPEATABILITY

Ask the customer to repeat the sample 3-5 times

Example 1: Target result is 5.42%

R1 = 5.56%

R1 = 5.53%

R1 = 5.50%

R1 = 5.47%

R1 = 5.43%

What's happening?

- Most likely, bubbles in the burette
- Possible Cross-contamination

Sources of Error



How to distinguish Systematic vs. Random?

Example 2: Target result is 5.42%

R1 = 5.54%

R1 = 5.53%

R1 = 5.54%

R1 = 5.54%

R1 = 5.53%

What's happening?

- Change in titer
- Inaccurate sampling
- Contaminated solvent
- The customer is wrong! it is supposed to be 5.54%

Sources of Error



How to distinguish Systematic vs. Random?

Example 3: Target result is 5.42%

R1 = 5.48%

R1 = NaN%

What's happening?

R1 = 5.48%

- Electrode failure

R1 = NaN%

R1 = NaN%

Customer also mentions the titrations take longer than they used to.

Agenda

- Sources of errors

Using the right titration method

- Reagent handling
- pH calibration and sensor care
- Sampling & weighing the right way
- Practicals



Using the right method for the task

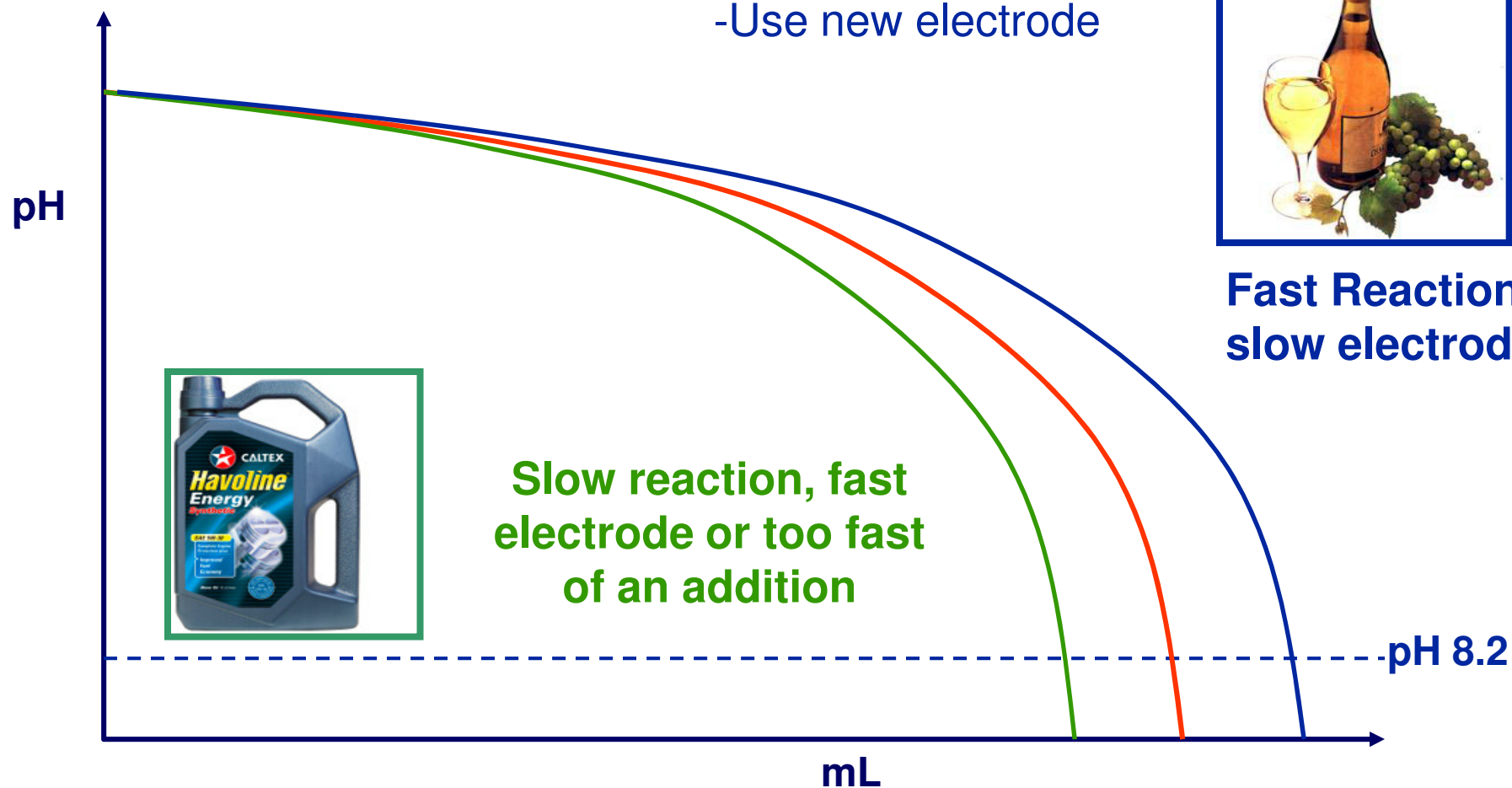


The end point titration

How could you confirm?

-Slow down titration

-Use new electrode



Using the right method for the task



The End Point Titration

- Often faster
- Approximation
- Electrode calibration
- Temperature
- Reaction speed
- Electrode response

Using the right method for the task



Two types of End Point Titration -> Absolute vs. Relative

- Often faster
- Approximation
- Electrode calibration
- Temperature
- Reaction speed
- Electrode response

Redox Titrations

All that is needed

Done with MEASURE or EST

No longer applicable

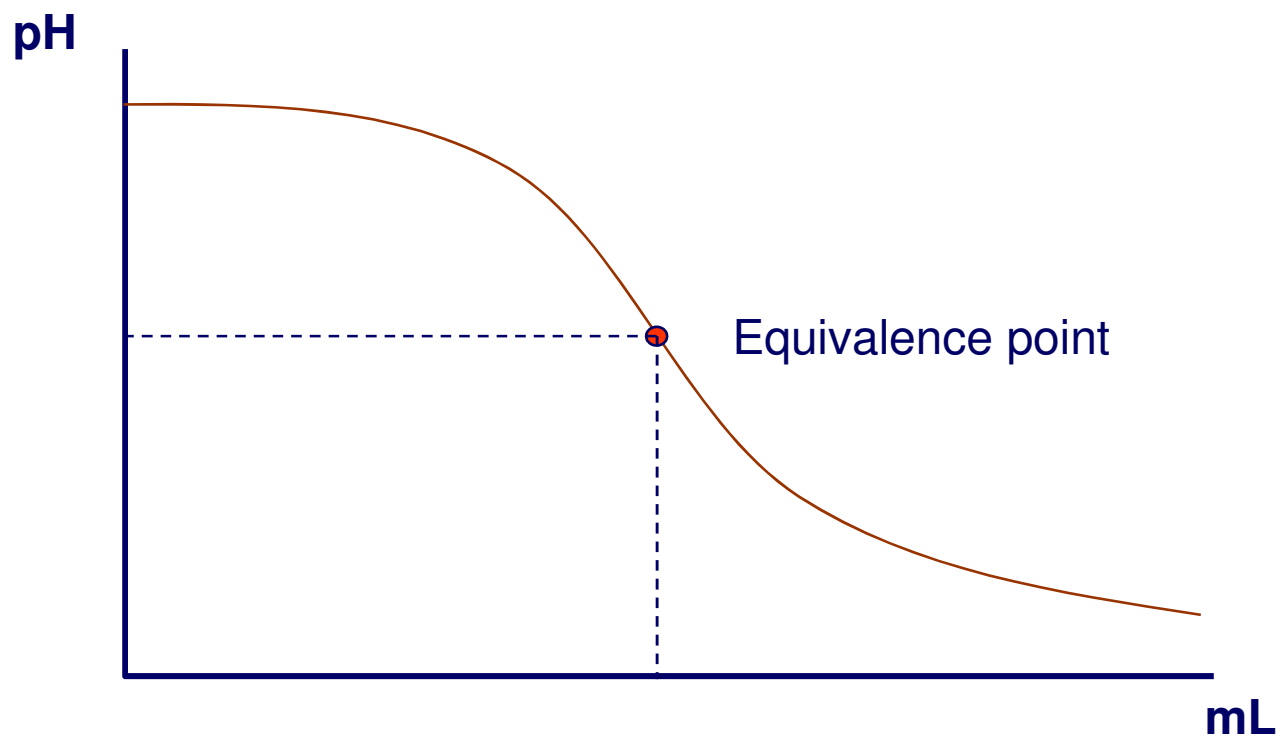
Dynamic concentration

Delay compensates

Using the right method for the task



The equivalence point titration



ERROR: stackunderflow
OFFENDING COMMAND: ~

STACK: